

The Positivity Wall: Geometric Routes to the Riemann Hypothesis, and Where They Stop

(VFD programme — expository working draft)

2026

Abstract

This is an *expository* synthesis. We recall how the Riemann Hypothesis (RH) reduces to a single positivity, survey the many equivalent forms this positivity takes (spectral, algebraic, dynamical, geometric), and explain why every geometric “route” we examined reaches the same inequality without crossing it. We then state, as precisely as the literature allows, the single object whose construction would close it—a canonical Frobenius over $\mathbb{Z}$ whose degree-form is the Weil positivity, i.e. the arithmetic surface of the function-field analogue. **No new result on RH is claimed.** Every mathematical statement below is standard and credited; the contribution is organizational: a clean map of the boundary, with explicit stamps separating what is known, what is verified here, and what is interpretation.

1 Scope

We claim no theorem about RH. The value of this note is a single, honestly stamped picture: RH is one inequality; it appears in many costumes; geometry stages it but does not close it; and the missing step is a construction, not a search. Readers seeking the verified arithmetic of the present programme are referred to the companion papers; here we only organize known mathematics.

2 RH as one positivity

By the Weil explicit-formula criterion [1], RH is equivalent to the nonnegativity of an explicit quadratic functional $W(f) \geq 0$ on a suitable space of test functions. Every route below is a re-expression of this one inequality.

3 One condition, many faces

The following are standard equivalent criteria and standard programme templates for RH. The first three are *proven equivalent* to RH; the last two are programme templates (one open over $\mathbb{Z}$, one a theorem in the function-field model). A single off-line zero falsifies the proven-equivalent criteria simultaneously:

- Weil positivity $W(f) \geq 0$ (spectral / Connes [2]) — equivalent to RH;
- the Li coefficients $\lambda_n \geq 0$ (algebraic [3]) — equivalent to RH;
- the de Bruijn–Newman constant $\Lambda \leq 0$ (dynamical; with $\Lambda \geq 0$ proven [5], so $\text{RH} \iff \Lambda = 0$);

- hyperbolicity of *all* Jensen polynomials of Ξ is equivalent to RH; GORZ [4] prove the fixed-degree/asymptotic part, not RH itself;
- the existence of a self-adjoint operator with the zeros as spectrum (Hilbert–Pólya) — a *programme*, not a proven equivalence;
- positivity of a Frobenius degree-form on a surface — a *theorem* in the function-field template [6, 7], with no built analogue over $\mathbb{Z}$.

Viewed as the proven-equivalent core, these are not many problems but one, seen from many sides; the last two are the geometric/spectral templates one hopes to transport.

4 The structure around the zeros (true, but not the wall)

The Riemann zeros are modelled, with strong numerical and partial-theorem support, by the statistics of a chaotic, time-reversal-broken (GUE, $\beta = 2$) spectrum: Montgomery [8] proves the pair correlation in a restricted range (conditionally), and Odlyzko [9] confirms it numerically to high precision; full GUE statistics remain conjectural. Summed as oscillations the zeros reproduce the primes with von-Mangoldt weights (the explicit formula, unconditional); and Rodgers–Tao give $\Lambda \geq 0$, so RH would force the de Bruijn–Newman constant to $\Lambda = 0$ (not known unconditionally). The explicit formula holds whether or not RH is true; the GUE statistics are a separate, largely empirical input. These constrain the shape of any solution; they do not supply the positivity.

5 The missing object

The function-field analogue of RH is a theorem precisely because the positivity is *geometric*: for a curve $C/\mathbb{F}_q$, the bound on Frobenius eigenvalues follows from the Hodge-index theorem on the surface $C \times C$. Over $\mathbb{Z}$ the analogous object—a canonical Frobenius whose degree-form is $W(f)$, living on a putative arithmetic surface $\text{Spec } \mathbb{Z} \times_{\mathbb{F}_1} \text{Spec } \mathbb{Z}$ —is *unbuilt*. The leading programme towards it—Connes’ trace-formula / noncommutative-geometry approach and the related $\mathbb{F}_1$ / absolute-geometry programmes (Connes–Consani [10])—recovers the explicit formula as a trace, but reduces RH to a *positivity of a trace pairing* that remains unproven. (Brave-new-ring heuristics model $\mathbb{F}_1$ by the sphere spectrum $\mathbb{S}$ and the surface by $\mathbb{Z} \otimes_{\mathbb{S}} \mathbb{Z}$; this is a separate suggestion, not part of Connes–Consani’s site.) The positivity is the wall.

6 A table of symbols, and which one is a placeholder

The RH sentence can be written entirely in standard notation; the notation is *complete*. What is incomplete is the *referent* of one symbol: it names a hoped-for object that has no agreed construction. We make this explicit.

symbol	reading	status
$\text{Re}(s) = \frac{1}{2}$	critical line	defined; computable
$ \alpha_i = q^{1/2}$	Frobenius eigenvalues on a circle	<i>theorem</i> (function field)
$\xi(s) = \xi(1-s)$	the mirror / functional equation	defined; proven
$W(f) \geq 0$	Weil positivity	defined; <i>unproven</i> (= RH)
Frob_q	geometric Frobenius	a real morphism
$\text{Spec } \mathbb{Z} \times_{\mathbb{F}_1} \text{Spec } \mathbb{Z}$	arithmetic surface	no construction carrying the required positivity
$\mathbb{F}_1$	“field with one element”	many frameworks; none supplies the RH object

Several inequivalent $\mathbb{F}_1$ -frameworks exist (§7); what none supplies is a canonical arithmetic surface carrying the required intersection theory and positivity. The reciprocal mirror $x \mapsto 1/x$ on $\mathbb{P}^1$ (radius $0 \leftrightarrow \infty$, the unit circle $|x| = 1$ fixed) is a schematic model of the involution behind $\xi(s) = \xi(1-s)$ under $s = \frac{1}{2} + it \leftrightarrow \log|x|$; that involution is built. The unbuilt entry is the $\mathbb{F}_1$ -surface. Writing the symbol is legitimate; *constructing the object it should point to is exactly the open problem.*

7 A ledger of $\mathbb{F}_1$ candidates

Because $\mathbb{F}_1$ is the placeholder, it is natural to ask whether one may simply *define* it—“ $\mathbb{F}_1 :=$ unity,” say—and proceed. Many definitions exist; the question is never whether it can be *named* but whether the named object *carries* the explicit-formula positivity *without assuming it*. We classify the principal published candidates by three documented criteria: DEFINED (an agreed object exists), CARRIES (building the surface over it yields $W(f)$ as a genuine degree/intersection form), and HONEST (it does so without putting the conclusion in by hand). The verdicts below report the literature’s status, not a computation: whether a given $\mathbb{F}_1$ proves RH *is* the open problem.

candidate	verdict	where it stops
$\mathbb{F}_1 = \{*\}$, trivial monoid (“unity”)	INERT	no nontrivial Frobenius
$\mathbb{F}_1$ -vector space = pointed set [11]	INERT	$q \rightarrow 1$ kills the arithmetic
Deitmar monoid schemes [12]	INERT	toric-like (finite-type) factors
Borger Λ -rings [13]	OPEN	supplies Frobenius lifts, not $W(f)$
Connes–Consani arithmetic site [10]	OPEN	trace recovered, positivity open
Deninger dynamical cohomology [14]	CONJECTURAL	cohomology hypothetical
“ $\mathbb{F}_1$ defined so $ \alpha = q^{1/2}$ ” (control)	CIRCULAR	assumes the conclusion

None of the listed candidates is known to supply the required positivity without an extra hypothesis. Every honestly-defined object is either INERT (real, but carries no zeros—the fate of “unity”), OPEN (Connes–Consani, Borger: reaches the wall, positivity not established), or CONJECTURAL (Deninger: the needed cohomology is hypothetical). The only candidate that *carries* the positivity outright is the CIRCULAR control, which assumes $|\alpha| = q^{1/2}$ and is rejected on the same provenance grounds that reject a least-squares fitted map. The honest summary: one can name a real object, or one that would cross; no listed framework is known to be both.

8 The joint void: a single door seen from two rooms (a conjectural picture)

This section is an interpretive picture, not a theorem. RH is approached from two sides that the Hilbert–Pólya philosophy and Connes’ programme *hope* are one object seen twice: an *arithmetic* side (a Frobenius on the $\mathbb{F}_1$ -surface whose degree-form would be $W(f) \geq 0$) and a *spectral* side (a self-adjoint $H = H^*$ whose spectrum would be the zeros—Hilbert–Pólya, Berry–Keating [15]). Tabulating what each side reaches today (“reach” = a partial construction exists; “part.” = modelled but not fully built; “dark” = no construction):

requirement	arithmetic	spectral	
explicit/trace formula	reach	reach	
global density (Weyl law)	reach	reach	
GUE local statistics (empirical)	part.	part.	shared floor (no RH)
mirror $\xi(s) = \xi(1-s)$	reach	reach	
Frobenius / arithmetic surface	part.	<i>dark</i>	geometry’s half (unbuilt)
self-adjoint H on a positive space	<i>dark</i>	part.	physics’ half (no final H)
$W(f) \geq 0$ / (spec H = zeros, $H = H^*$)	<i>dark</i>	<i>dark</i>	the door

Two observations. First, the two sides *already meet* on the structure that does not require RH (the shared floor; GUE only empirically). Second, the programme’s *hope* is that the arithmetic-dark and spectral-dark requirements are the same object missing its two faces. We emphasise the status carefully: Connes’ analysis shows RH is equivalent to the positivity of a *trace pairing*; it does *not* prove that “ $W(f) \geq 0$ ” and “ $H = H^*$ ” are literally the same statement, and **no theorem identifies the arithmetic and spectral constructions**. What can be said honestly is that *both, under their own strong hypotheses, would yield RH*, and the programme conjectures a single underlying object. If that conjecture held, the “both” requirement would not widen the problem but pin the missing object toward one shape—a heuristic motivation, not a lever. It does not lower the wall: supplying the positivity by fiat on either face is circular, as above. The honest gain is only a sharper *picture*—“a single door, seen from two rooms”—with the floor on both sides and each doorpost, at most, half-built.

9 Closing

Every accessible route—cavity, mirror, operator, loop, gas, flow, hyperbolicity—is a re-expression of one positivity, and re-expression does not prove it. The single missing step is the construction of an arithmetic surface and the positivity of its intersection form. We make no claim of progress on that step; we have only drawn the boundary cleanly, and marked, at each face, exactly where known mathematics ends.

A note on interpretation. Several physically suggestive readings arise naturally when one draws these pictures (curvature, mirrors, resonance). They are *interpretation only*; no statement in this note depends on them, and none is claimed as mathematics.

References

- [1] A. Weil, *Sur les “formules explicites” de la théorie des nombres premiers*.
- [2] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (1999).
- [3] X.-J. Li, *The positivity of a sequence of numbers and the Riemann Hypothesis*, J. Number Theory (1997).
- [4] M. Griffin, K. Ono, L. Rolin, D. Zagier, *Jensen polynomials for the Riemann zeta function and other sequences*, PNAS (2019).
- [5] B. Rodgers, T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum Math. Pi (2020).
- [6] A. Weil, *Sur les courbes algébriques et les variétés qui s’en déduisent* (1948).

- [7] P. Deligne, *La conjecture de Weil I*, IHES (1974).
- [8] H. Montgomery, *The pair correlation of zeros of the zeta function* (1973).
- [9] A. Odlyzko, *On the distribution of spacings between zeros of the zeta function* (1987).
- [10] A. Connes, C. Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*; and subsequent work on the arithmetic/scaling site.
- [11] H. Cohn, *Projective geometry over $\mathbb{F}_1$ and the Gaussian binomial coefficients*, Amer. Math. Monthly (2004).
- [12] A. Deitmar, *Schemes over $\mathbb{F}_1$* , in *Number Fields and Function Fields* (2005); see also B. Toën, M. Vaquié, *Au-dessous de $\text{Spec } \mathbb{Z}$* (2009).
- [13] J. Borger, *Λ -rings and the field with one element* (2009).
- [14] C. Deninger, *Some analogies between number theory and dynamical systems on foliated spaces*, ICM (1998).
- [15] M. V. Berry, J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Review (1999).